

The current generation of processors have up to eight cores and most smartphone users are gradually shifting to octa-core SoC-powered devices.

With the advent of graphically intensive games, SoC developers have also been forced to include a high quality GPU in the chips, which would be able to run these games without any major lags.

The other important feature which determines the performance of an SoC is its architecture and process count. For instance, most SoCs in the latest devices come with a 14nm process. However, there are some products which have been developed with a better 10nm process. These are references to the transistors which are housed in the chips.

What Is Transistor Count In SoC?

Transistor count refers to the number of transistors present in the integrated circuit of the SoC. The higher the transistor count, the more complex the circuitry is. According to Moore's Law, the transistor count for integrated circuits doubles after every two years.

The law was developed by Intel's co-founder Gordon Moore in 1965, when he identified a pattern in the development of ICs. He noticed that transistor count doubled every two years. So, the complexity of these circuits too increased twofold every two years.

In 2015, Intel became the first company to create a chip which housed over 5.5 billion transistors. It is as of yet, the highest transistor count in a single chip. However, with the exponential increase in transistor counts, new and updated technology has to be integrated into the chips. This has resulted in the rise of FinFET technology.

Why Size Reduction Of Process Is Required?

To include even greater transistor counts for faster and more efficient chips, companies have been pushing the current technology. In a bid to include more transistors, developers needed to create extra space in processors.

This is why process size has to be reduced so as to create more space for transistors, which would further improve the overall performance.

What Is FinFET?

FinFET refers to Fin Field Effect Transistor which replaces the 2D transistors with 3D ones. This technology was adapted in recent years after developers faced problems in integrating the growing number of transistors to the chips. To make the chips smaller, developers scaled its